MTH 200 Chapter 4: Applications of Derivatives

4.2 Linear Approximations & Differentials Worksheet

1. Find the linearization of the function $f(x)=\sqrt{x+3}$ at $x=1$.

   If $x=1$, then $f(1)=\sqrt{1+3}=2$, so we have the point: $(1,2)$.

   The slope of the line is given by $f'(1)$. We see that $f'(x)=\dfrac{1}{2\sqrt{x+3}}$. Therefore, $f'(1)=\dfrac{1}{2\sqrt{4}}=\dfrac{1}{4}$.

   So the *linearization at* $x=1$ is:

   $$L(x)=f'(1)(x-1)+f(1) \quad\rightarrow\quad L(x)=\frac{1}{4}(x-1)+2$$

   We know that $f(x)\approx L(x)$ for values of x near 1.

2. Find the linearization of the function $f(x)=\ln(2x-3)$ at $x=2$.

   If $x=2$, then $f(2)=\ln(2(2)-3)=0$, so we have the point: $(2,0)$.

   The slope of the line is given by $f'(2)$. We see that $f'(x)=\dfrac{2}{2x-3}$. Therefore, $f'(2)=\dfrac{2}{2(2)-3}=2$.

   So the *linearization at* $x=2$ is:

   $$L(x)=f'(2)(x-2)+f(2) \quad\rightarrow\quad L(x)=2(x-2)+0$$

   We know that $f(x)\approx L(x)$ for values of x near 2.

3. Use a linearization to approximate the number $\sqrt{3.98}$.

   $\sqrt{3.98}$ is hard to calculate by hand, but $\sqrt{4}$ is easy. Let $f(x)=\sqrt{x}$.

   We will make a linearization at $x=4$ and then estimate $\sqrt{3.98}=f(3.98)$.

   If $x=4$, then $f(4)=\sqrt{4}=2$, so we have the point: $(4,2)$.

   The slope of the line is given by $f'(4)$. We see that $f'(x)=\dfrac{1}{2\sqrt{x}}$. Therefore, $f'(4)=\dfrac{1}{2\sqrt{4}}=\dfrac{1}{4}$.

   So the *linearization at* $x=4$ is:

   $$L(x)=f'(4)(x-4)+f(4) \quad\rightarrow\quad L(x)=\frac{1}{4}(x-4)+2$$

   We know that $f(x)\approx L(x)$ for values of x near 4. Therefore,

   $$f(3.98)\approx L(3.98)=\frac{1}{4}(3.98-4)+2=\frac{1}{4}(-0.02)+2=-0.005+2=1.995$$

4. Approximate the number $(1.06)^6$.

$(1.06)^6$ is hard to calculate by hand, but $(1)^6$ is easy. Let $f(x)=x^6$.

We will make a linearization at $x=1$ and then estimate $(1.06)^6=f(1.06)$.

If $x=1$, then $f(1)=(1)^6=1$, so we have the point: $(1,1)$.

The slope of the line is given by $f'(1)$. We see that $f'(x)=6x^5$. Therefore, $f'(1)=6(1)^5=6$.

So the *linearization at* $x=1$ is:

$$L(x)=f'(1)(x-1)+f(1) \quad \to \quad L(x)=6(x-1)+1$$

We know that $f(x)\approx L(x)$ for values of x near 1. Therefore,

$$f(1.06)\approx L(1.06)=6(1.06-1)+1=6(0.06)+1=0.36+1=1.36$$

5. Use the techniques of linear approximation to estimate the following quantities. Write each answer in decimal form. *Show all work by hand.*

a) $\ln(0.9)$

$\ln(0.9)$ is hard to calculate by hand, but $\ln(1)$ is easy. Let $f(x)=\ln(x)$.

We will make a linearization at $x=1$ and then estimate $\ln(1)=f(1)$.

If $x=1$, then $f(1)=\ln(1)=0$, so we have the point: $(1,0)$.

The slope of the line is given by $f'(1)$. We see that $f'(x)=\frac{1}{x}$. Therefore, $f'(1)=\frac{1}{(1)}=1$.

So the *linearization at* $x=1$ is:

$$L(x)=f'(1)(x-1)+f(1) \quad \to \quad L(x)=1(x-1)+0$$

We know that $f(x)\approx L(x)$ for values of x near 1. Therefore,

$$f(0.9)\approx L(0.9)=1(0.9-1)+0=-0.1$$

b) $\sqrt{104}$

$\sqrt{104}$ is hard to calculate by hand, but $\sqrt{100}$ is easy. Let $f(x)=\sqrt{x}$.

We will make a linearization at $x=100$ and then estimate $\sqrt{104}=f(104)$.

If $x=100$, then $f(100)=\sqrt{100}=10$, so we have the point: $(100,10)$.

The slope is given by $f'(100)$. We see that $f'(x)=\frac{1}{2\sqrt{x}}$. Therefore, $f'(100)=\frac{1}{2\sqrt{100}}=\frac{1}{20}$.

So the *linearization at* $x=100$ is:

$$L(x)=f'(100)(x-100)+f(100) \quad \to \quad L(x)=\frac{1}{20}(x-100)+10$$

We know that $f(x)\approx L(x)$ for values of x near 100. Therefore,

$$f(104)\approx L(104)=\frac{1}{20}(104-100)+10=\frac{1}{20}(4)+10=0.2+10=10.2$$

c) $\sin\left(\frac{\pi}{4}+0.02\right)$

$\sin\left(\frac{\pi}{4}+0.02\right)$ is hard to calculate by hand, but $\sin\left(\frac{\pi}{4}\right)$ is easy. Let $f(x)=\sin(x)$.

We will make a linearization at $x=\frac{\pi}{4}$ and then estimate $\sin\left(\frac{\pi}{4}+0.02\right)=f\left(\frac{\pi}{4}+0.02\right)$.

If $x=\frac{\pi}{4}$, then $f\left(\frac{\pi}{4}\right)=\sin\left(\frac{\pi}{4}\right)=\frac{\sqrt{2}}{2}$, so we have the point: $\left(\frac{\pi}{4},\frac{\sqrt{2}}{2}\right)$.

The slope is given by $f'\left(\frac{\pi}{4}\right)$. We see that $f'(x)=\cos(x)$. Therefore, $f'\left(\frac{\pi}{4}\right)=\cos\left(\frac{\pi}{4}\right)=\frac{\sqrt{2}}{2}$.

So the *linearization at* $x=\frac{\pi}{4}$ is:

$$L(x)=f'\left(\frac{\pi}{4}\right)\left(x-\frac{\pi}{4}\right)+f\left(\frac{\pi}{4}\right) \quad\rightarrow\quad L(x)=\frac{\sqrt{2}}{2}\left(x-\frac{\pi}{4}\right)+\frac{\sqrt{2}}{2}$$

We know that $f(x)\approx L(x)$ for values of x near $\frac{\pi}{4}$. Therefore,

$$f\left(\frac{\pi}{4}+0.02\right)\approx L\left(\frac{\pi}{4}+0.02\right)=\frac{\sqrt{2}}{2}\left(\frac{\pi}{4}+0.02-\frac{\pi}{4}\right)+\frac{\sqrt{2}}{2}=0.02\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}=1.02\frac{\sqrt{2}}{2}=0.51\sqrt{2}$$

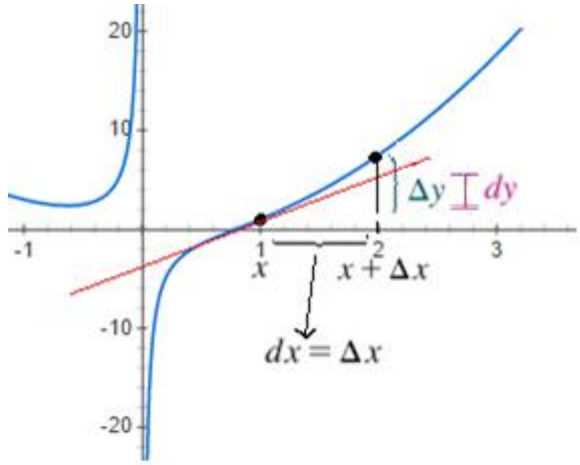
6. Consider the function $y=2x^2-\frac{1}{x}$.

a) Approximate Δy and dy as x changes from 1 to 2.

$$\Delta y=f(2)-f(1)=7.5-1=6.5$$

$$\frac{dy}{dx}=4x+\frac{1}{x^2} \quad\rightarrow\quad dy=\left(4x+\frac{1}{x^2}\right)dx \quad\rightarrow\quad dy=\left(4(1)+\frac{1}{(1)^2}\right)(1)=5$$

b) Sketch a picture of the curve and the values of Δy and dy.



7. Consider the function $y = x\sqrt{8x+1}$.

a) Find general formulas for Δy and dy.

$$\Delta y = y(x+\Delta x) - y(x) = (x+\Delta x)\sqrt{8(x+\Delta x)+1} - x\sqrt{8x-1}$$

$$\frac{dy}{dx} = x\frac{1}{2}(8x+1)^{-1/2}(8) + \sqrt{8x+1} \quad \rightarrow \quad dy = \left[\frac{4x}{\sqrt{8x+1}} + \sqrt{8x+1}\right]dx$$

b) Approximate Δy and dy when $x = 3$ and $\Delta x = 0.01$.

$$\Delta y = f(3.01) - f(3) \approx 15.07 - 15 = 0.07$$

$$dy = \left[\frac{4(3)}{\sqrt{8(3)+1}} + \sqrt{8(3)+1}\right](0.01) = \left(\frac{12}{5} + 5\right)(0.01) = 0.074$$